

Microeconomics II – Class 10

Slutsky equation



Class summary

- * Slutsky equation
 - * Slutsky's substitution effect
 - * Income effect
 - * Slutsky's identity
 - * Sign of the substitution and income effects
 - * Inferior goods and Giffen goods
- * Readings: Varian (chap. 8 and 6.1-6.4)



Microeconomics I review

- * Today's goal: To analyse the changes in demand caused by the changes in prices.
- * 2 possible demand reactions
 - * For normal goods (and many inferior goods, too)

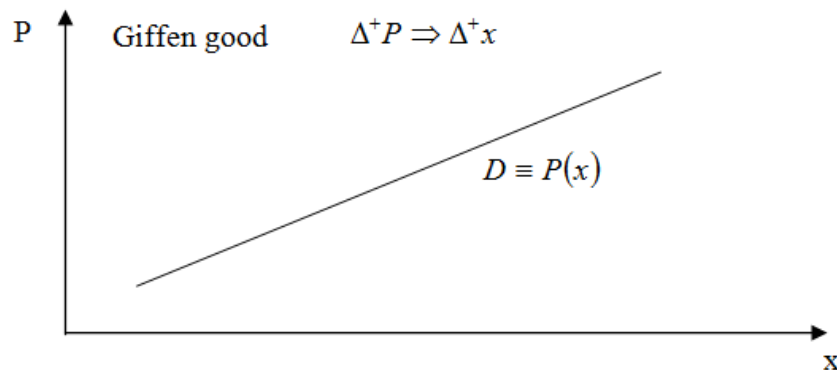
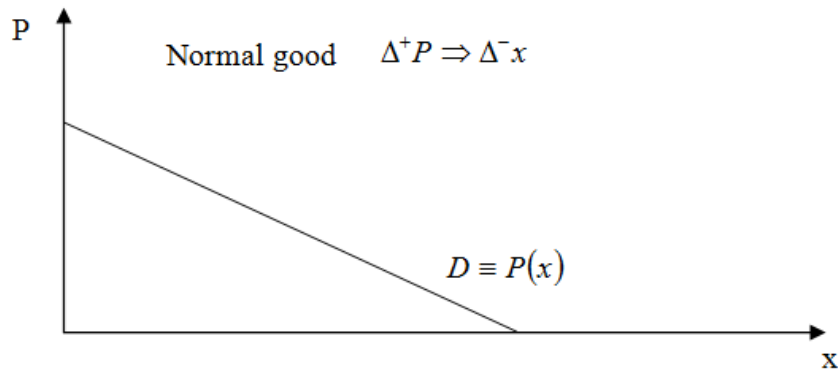
$$\Delta^+ P \Rightarrow \Delta^- x$$

- * For Giffen goods

$$\Delta^+ P \Rightarrow \Delta^+ x$$



Microeconomics I review



Microeconomics I review

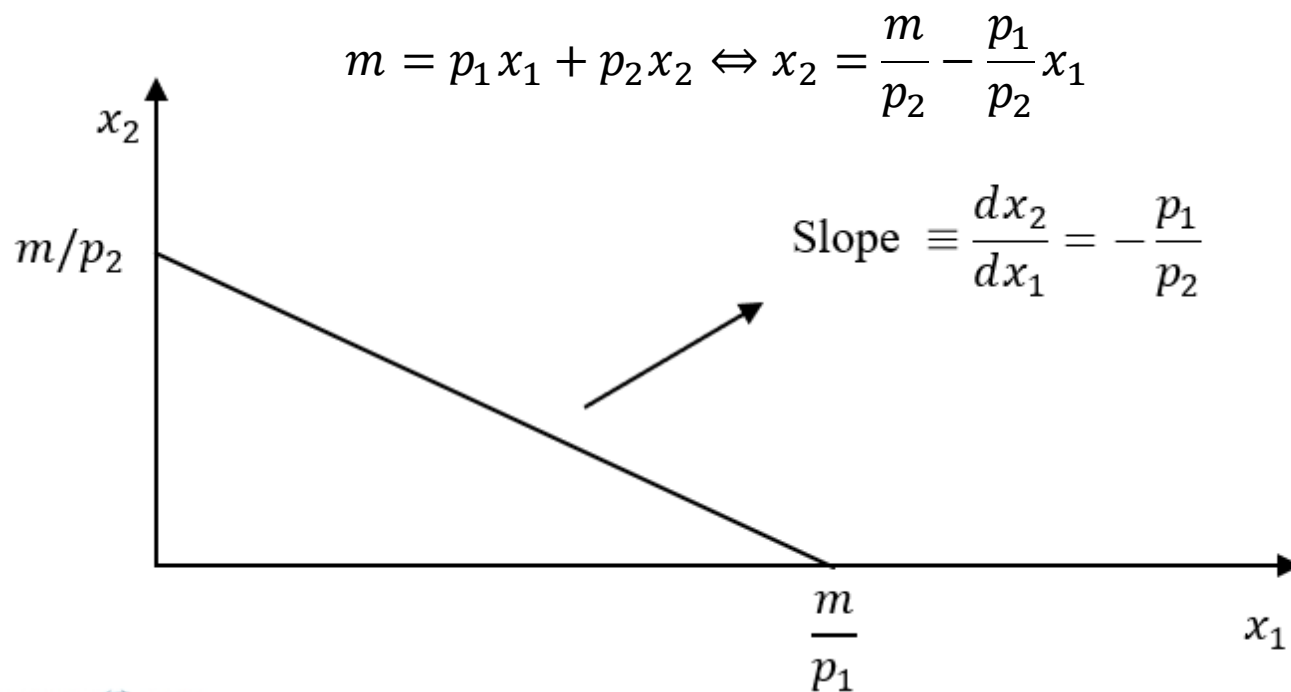
- * Giffen goods are rare.
(Hopefully, by the end of the class you'll know why)
- * We can decompose the change in demand (total effect) caused by a change in the price of a good into 2 effects:
 - * Substitution effect
 - * Income effect

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n$$



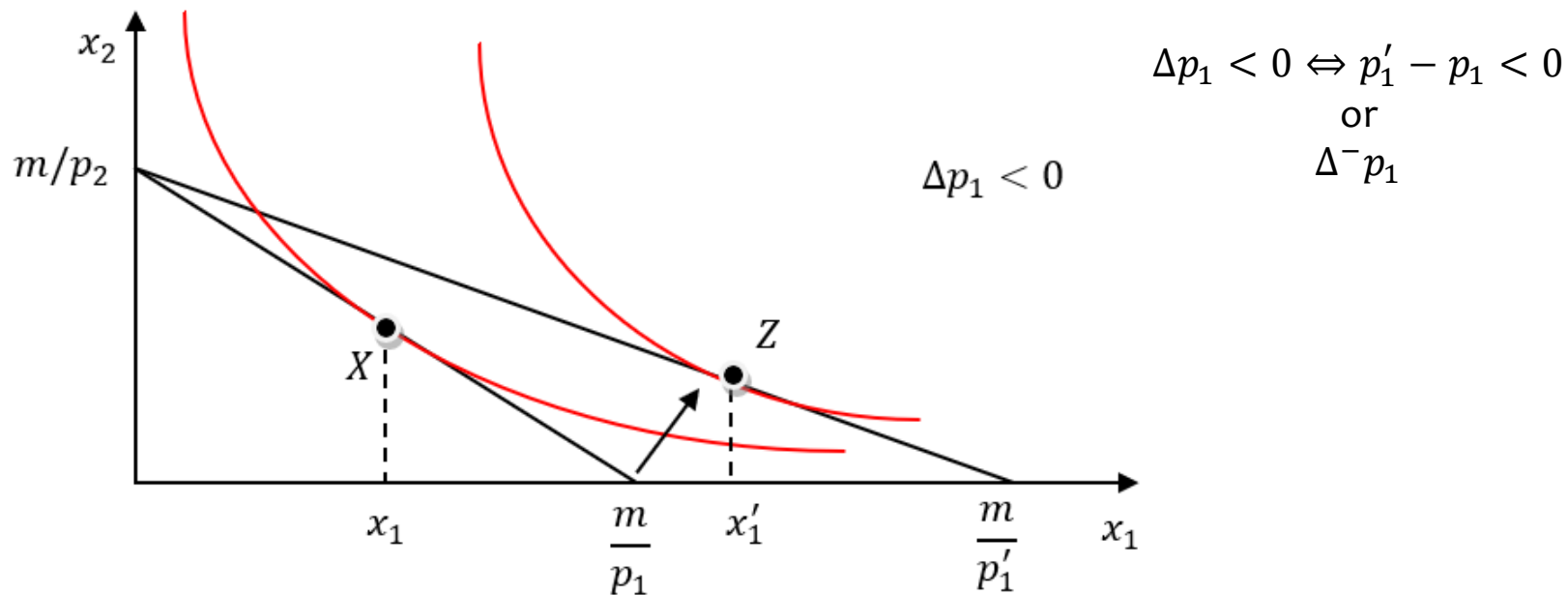
Slutsky's substitution effect

* Budget constraint



Slutsky's substitution effect

* Change in price (ex.: decrease)

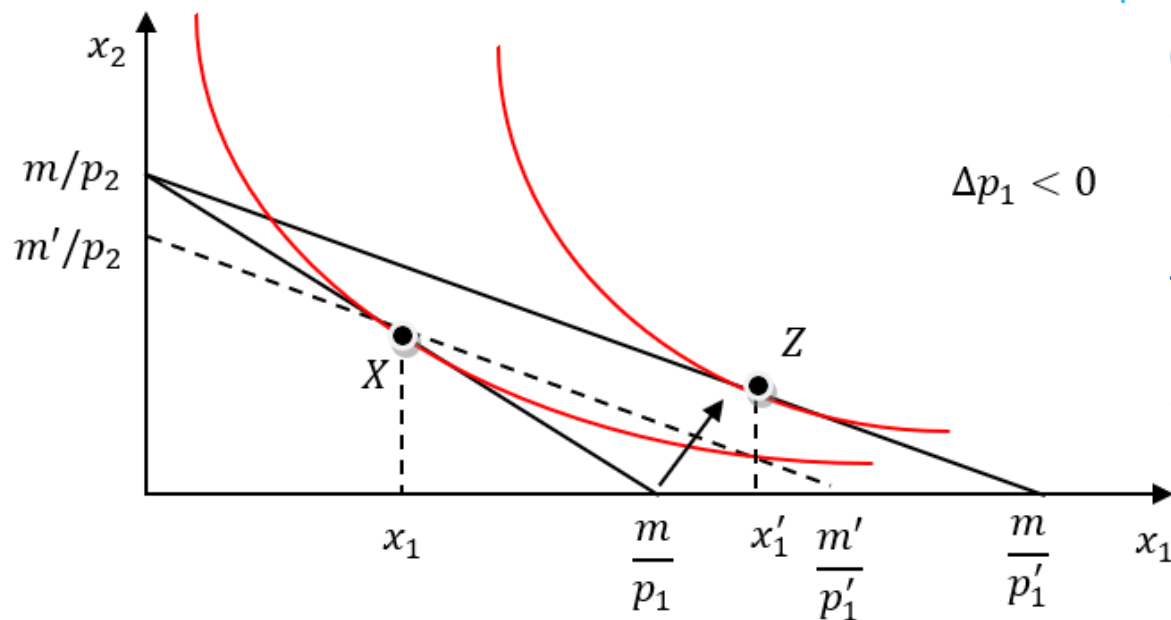


$\Delta x_1 = x'_1 - x_1 \equiv \text{Total effect on demand of the change in price of good 1}$



Slutsky's substitution effect

$$m' = p'_1 x_1 + p_2 x_2$$



- * For which value of income (m') would the previously chosen consumption bundle remain on the budget constraint despite the change in price, keeping purchasing power constant?



Slutsky's substitution effect

- * Change in income required to keep the consumer's purchasing power constant:

$$\begin{aligned} m' - m &= (p'_1 x_1 + p_2 x_2) - (p_1 x_1 + p_2 x_2) \Leftrightarrow \\ \Leftrightarrow \Delta m &= (p'_1 x_1 - p_1 x_1) + (p_2 x_2 - p_2 x_2) \Leftrightarrow \Delta m = (p'_1 - p_1) x_1 \Leftrightarrow \\ &\Leftrightarrow \Delta m = \Delta p_1 x_1 \end{aligned}$$

Ex.: (adapted from Varian (2010), p. 139, with some changes):

$p_1 \equiv$ Price for one bus ticket; $p_1 = €0.5$

We know that a student buys 20 tickets per week

$$x_1 = 20$$

If the price of bus tickets increases to $p_1 = €0.6$, how much would this student's weekly allowance have to increase for him to afford the same 20 weekly trips?



Slutsky's substitution effect

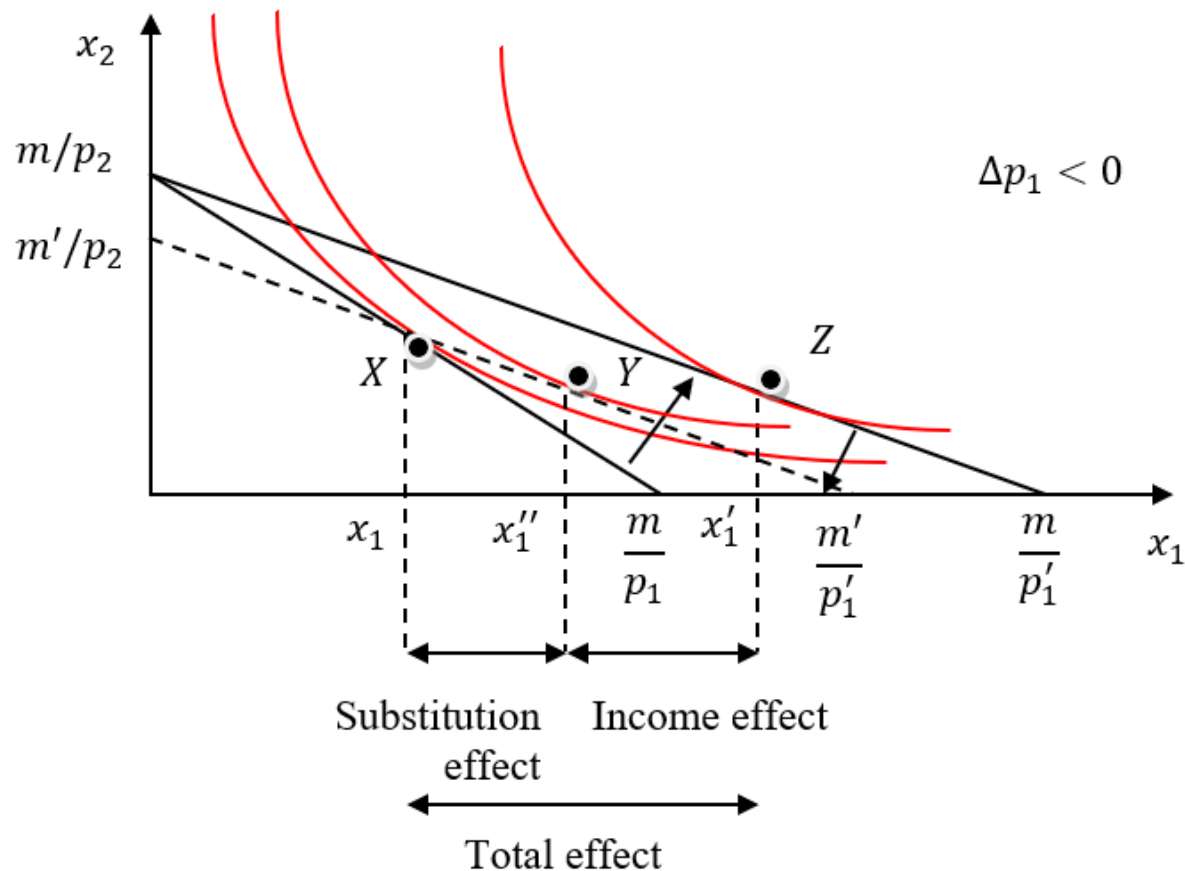
- * Substitution effect

- * The substitution effect measures the change in demand for a good caused by a change in the price of that good, keeping purchasing power ($\neq$ income) constant; it measures the impact of the change in relative prices (p_1/p_2).

$$\Delta x_1^s = x_1(p'_1, m') - x_1(p_1, m)$$



Slutsky's substitution effect



Slutsky's substitution effect

Example :

$$x_1 = 10 + \frac{m}{10p_1}$$

$$m = 120$$

$$p_1 = 3$$

$$p'_1 = 2$$

$$\Delta x_1 ? \Delta m ? m' ? \Delta x_1^s ?$$

- * How to compute the substitution effect from the demand function
- * How does the quantity demanded change?

$$\Delta x_1 = 16 - 14 = 2$$

- * How much income is needed to keep purchasing power constant?

$$\Delta m = \Delta p_1 x_1 \Leftrightarrow \Delta m = (2 - 3) \times 14 \Leftrightarrow \Delta m = -14$$



Slutsky's substitution effect

Example :

$$x_1 = 10 + \frac{m}{10p_1}$$

$$m = 120$$

$$p_1 = 3$$

$$p'_1 = 2$$

$$\Delta x_1 ? \Delta m ? m' ? \Delta x_1^s ?$$

* What is the amount of the substitution effect?

$$\Delta x_1^s = x_1(p'_1, m') - x_1(p_1, m) =$$

$$= x_1(2, 106) - x_1(3, 120) =$$

$$= \left(10 + \frac{106}{10 \times 2}\right) - \left(10 + \frac{120}{10 \times 3}\right) =$$

$$= 15.3 - 14 = 1.3$$



Income effect

- * Income effect
- * difference between the total effect and the substitution effect

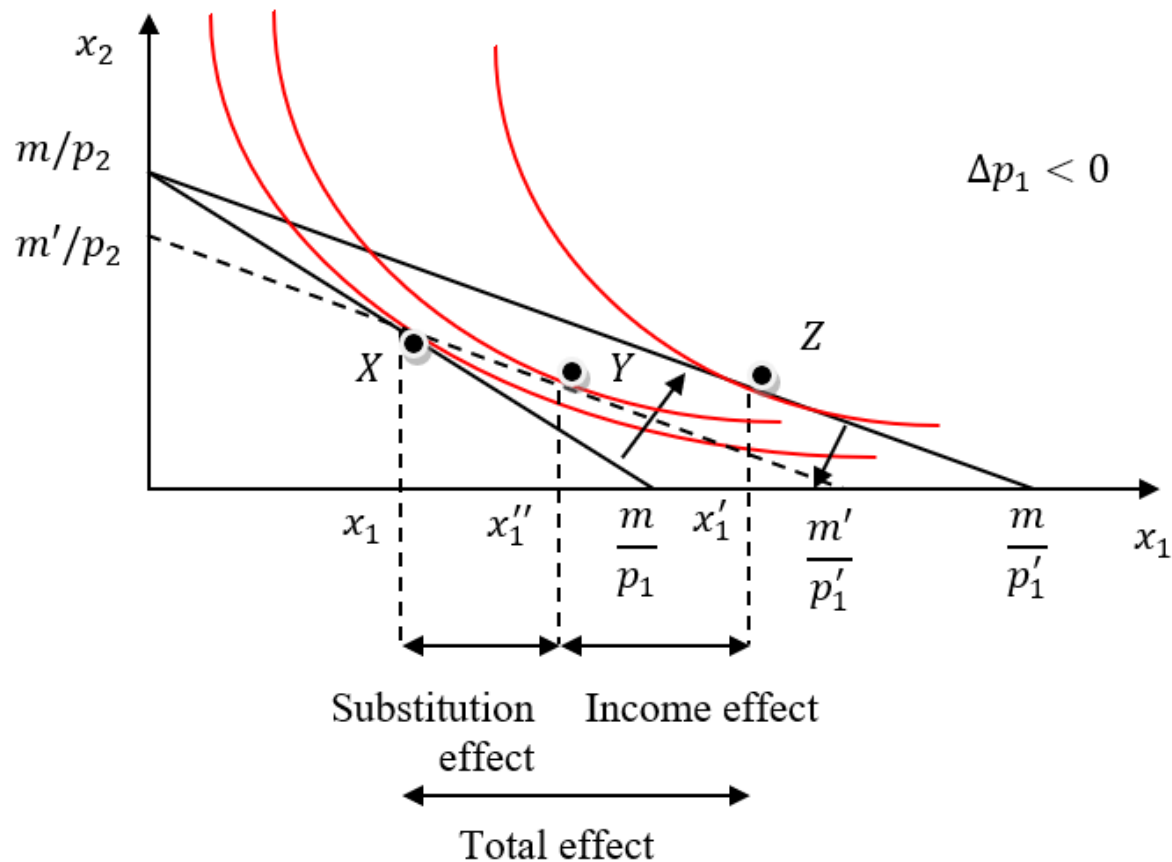
$$\text{Total effect: } \Delta x_1 = x_1(p'_1, m) - x_1(p_1, m)$$

$$\text{Substitution effect: } \Delta x_1^s = x_1(p'_1, m') - x_1(p_1, m)$$

$$\text{Income effect: } \Delta x_1^n = x_1(p'_1, m) - x_1(p'_1, m')$$



Income effect



Slutsky's identity

- * Total effect = substitution effect + income effect

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n$$

- * We can find out if demand increases or decreases as a result of a price change if we know the size and signal of the substitution and income effect.



Sign of the income effect

* Sign of the income effect

Normal good: $\Delta p_1 < 0 \Rightarrow \Delta^+ \text{ purchasing power} \Rightarrow \Delta x_1^n > 0$

Inferior good: $\Delta p_1 < 0 \Rightarrow \Delta^+ \text{ purchasing power} \Rightarrow \Delta x_1^n < 0$

Normal good: $\Delta p_1 > 0 \Rightarrow \Delta^- \text{ purchasing power} \Rightarrow \Delta x_1^n < 0$

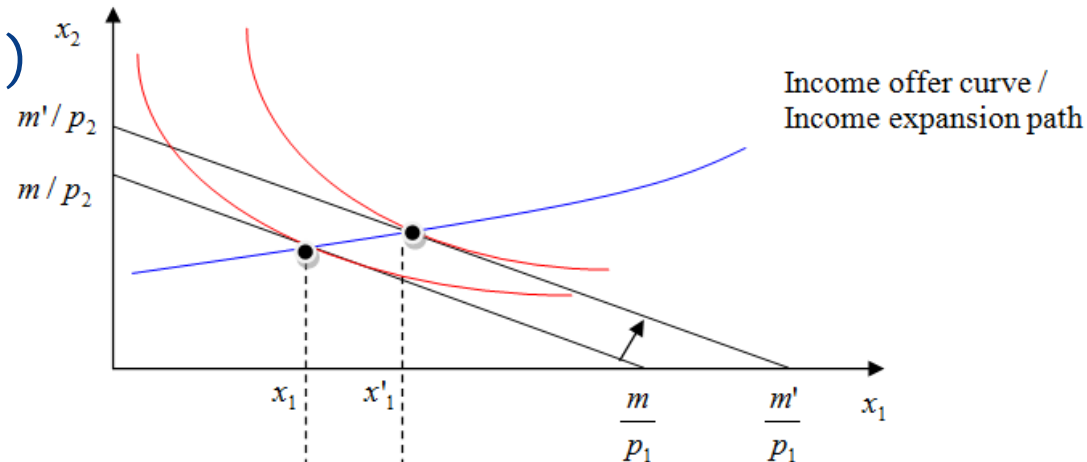
Inferior good: $\Delta p_1 > 0 \Rightarrow \Delta^- \text{ purchasing power} \Rightarrow \Delta x_1^n > 0$



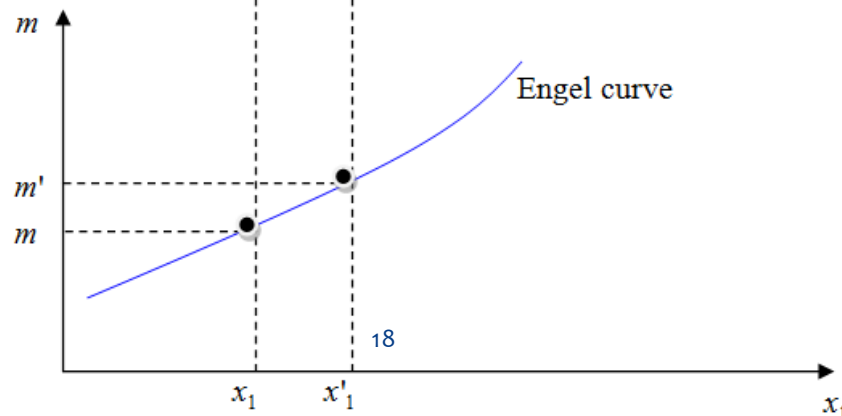
Microeconomics I review

* Normal good (x_1)

**Income offer curve /
Income expansion path:**
Set of optimal consumption bundles for the different income levels



Engel curve:
Relationship between income level and the demand for a given good

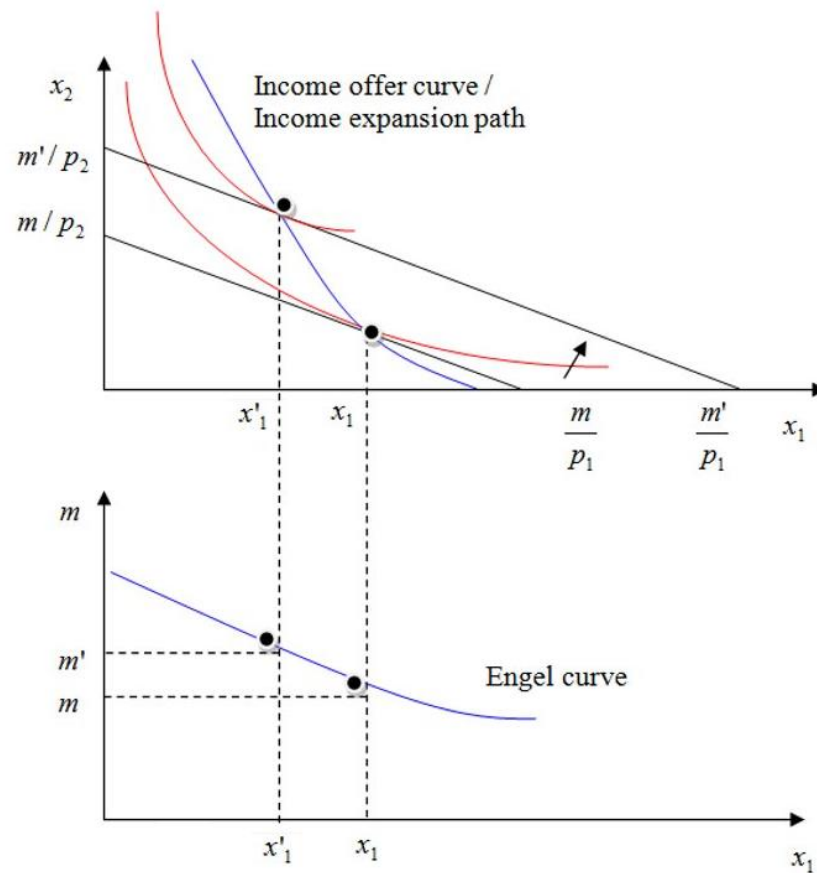


Microeconomics I review

* Inferior good (x_1)

**Income offer curve /
Income expansion path:**
Set of optimal consumption
bundles for the different
income levels

Engel curve:
Relationship between
income level and the
demand for a given good



Microeconomics I review

* Inferior good

$$\varepsilon_{x_1,m} < 0$$

* Normal good

$$\varepsilon_{x_1,m} \geq 0$$

* Necessary good

$$0 \leq \varepsilon_{x_1,m} \leq 1$$

* Luxury good

$$1 < \varepsilon_{x_1,m}$$

Income-elasticity of demand

$$\varepsilon_{x_1,m} \equiv \frac{\partial x_1}{\partial m} \times \frac{m}{x_1}$$



Income effect (example)

Example :

$$x_1 = 10 + \frac{m}{10p_1}$$

$$m = 120$$

$$p_1 = 3$$

$$p_1' = 2$$

$$\Delta x_1^n ?$$

- * Calculate the income effect using the demand function

$$\Delta x_1^n = x_1(p_1', m) - x_1(p_1', m')$$

$$= x_1(2, 120) - x_1(2, 106) =$$

$$= \left(10 + \frac{120}{10 \times 2}\right) - \left(10 + \frac{106}{10 \times 2}\right) =$$

$$= 16 - 15.3 = 0.7$$



Income effect (example)

Example :

$$x_1 = 10 + \frac{m}{10p_1}$$

$$m = 120$$

$$p_1 = 3$$

$$p'_1 = 2$$

$$\Delta x_1^n ?$$

* We have decomposed the total effect into substitution effect and income effect for this example:

$$\text{Total effect: } \Delta x_1 = x_1(p'_1, m) - x_1(p_1, m) = 2$$

$$\text{Substitution effect: } \Delta x_1^s = x_1(p'_1, m') - x_1(p_1, m) = 1.3$$

$$\text{Income effect: } \Delta x_1^n = x_1(p'_1, m) - x_1(p'_1, m') = 0.7$$



Sign of the substitution effect

- * The substitution effect always has a sign opposite to the sign of the price change.
- * Varian (2014) says it is “negative” but beware (!) it can have any sign!

$$\Delta p_1 < 0 \Rightarrow \Delta x_1^s \geq 0$$

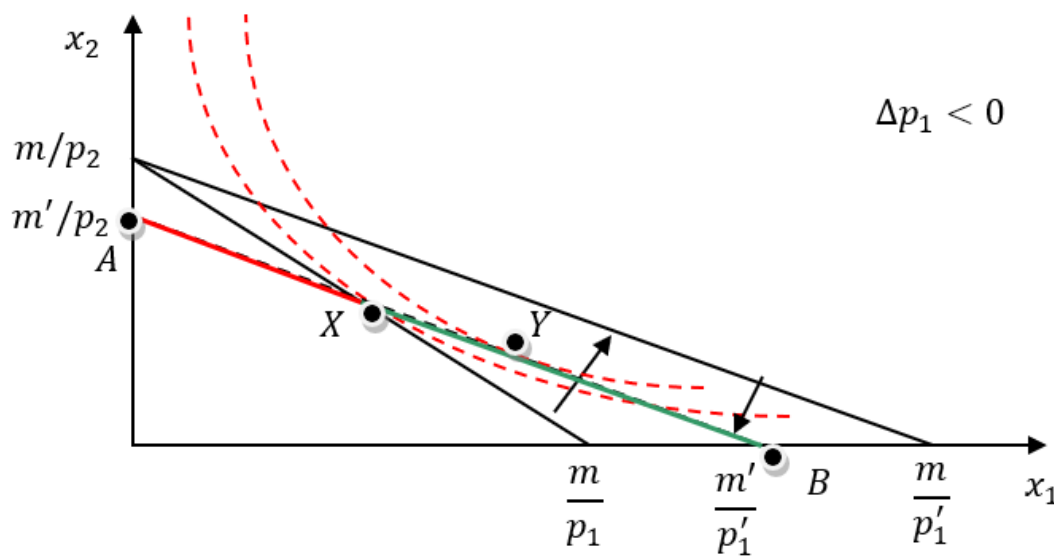
$$\Delta p_1 > 0 \Rightarrow \Delta x_1^s \leq 0$$



Sign of the substitution effect

* Proof using the weak axiom of revealed preference

For $\Delta^- p_1$ (do it at home for a price increase)



The bundle X has been revealed preferred to all bundles on the line segment $[AX[$ and is still affordable after the price change, therefore, the new chosen bundle for (p'_1, m_1) will always lie on the line segment $[XB]$.

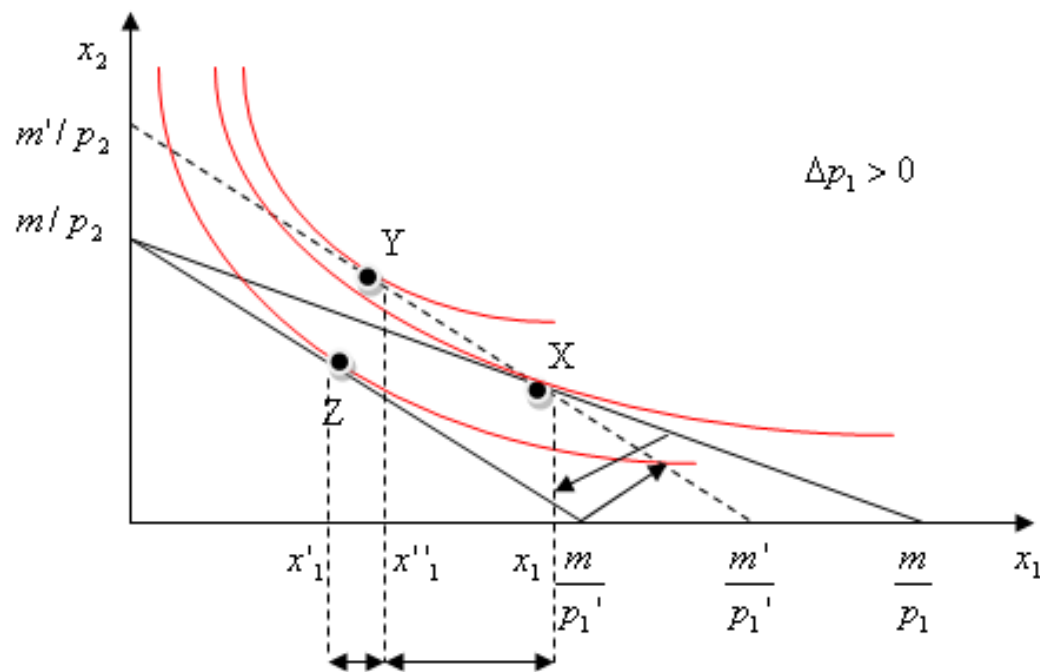
For every bundle Z in $[AX[$, we have that $X \succ Z$, therefore we cannot have $Z \succ X$ if the weak axiom of revealed preference holds.



Sign of the total effect

* For normal goods

$$\Delta p_1 > 0 \Rightarrow \begin{cases} \Delta x_1^s < 0 \\ \Delta x_1^n < 0 \end{cases} \Rightarrow \Delta x_1 < 0$$



Income effect Substitution effect
 $\Delta x_1^n < 0$ $\Delta x_1^s < 0$



Total effect
 $\Delta x_1 < 0$



Sign of the total effect

* Inferior goods

$$\Delta p_1 > 0 \Rightarrow \begin{cases} \Delta x_1^s < 0 \\ \Delta x_1^n > 0 \end{cases} \Rightarrow \Delta x_1 ?$$

$$|\Delta x_1^n| > |\Delta x_1^s| \Rightarrow \Delta x_1 > 0$$

$$|\Delta x_1^n| < |\Delta x_1^s| \Rightarrow \Delta x_1 < 0$$

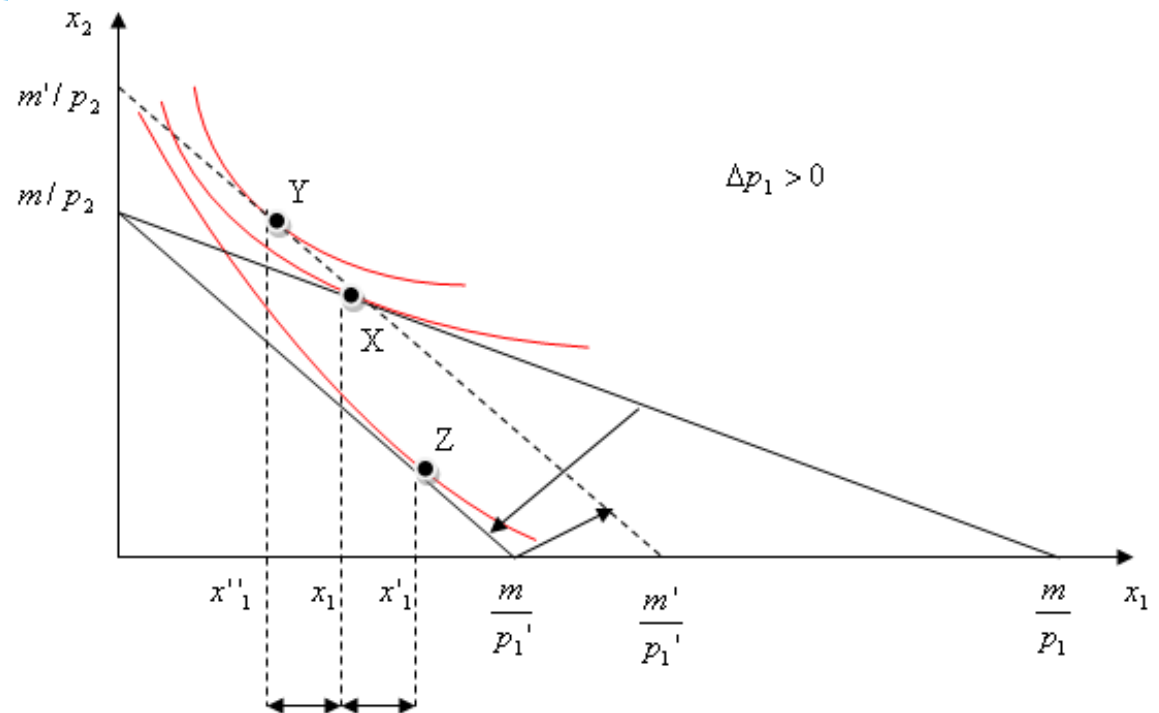
Giffen good

As for normal good



Sign of the total effect

* Giffen goods

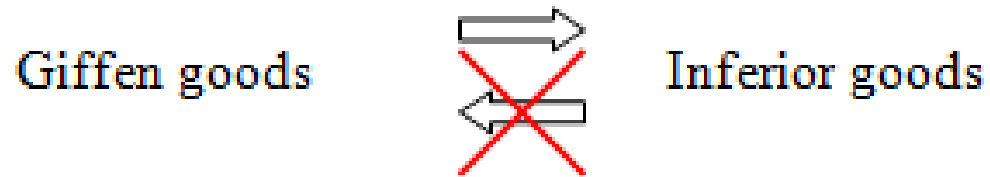


Substitution effect Total effect
 $\Delta x_1^s < 0$ $\Delta x_1 > 0$
 (Giffen good)

Income effect
 $\Delta x_1^i > 0$ (Inferior good)



Giffen goods and inferior goods



This is why Giffen goods are rare!



Giffen goods and Veblen goods

- * While a Giffen goods must be an inferior good, another type of good has demand increasing with price
- * Veblen goods are luxury goods used for conspicuous consumption, for consumers to signal status
- * We would need a broader framework to study this

